4Corruption Dataset Detection

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Outline

- Dataset description
- Proposed Framework
- Experiments
- Conclusion

Dataset-4Corruption

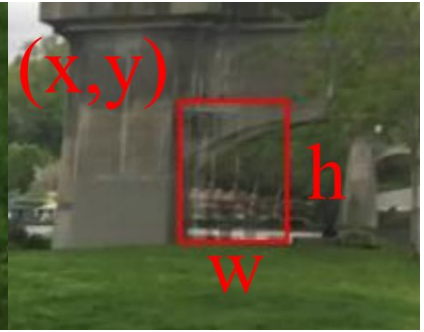
Total Videos: 50

- Frames per Video: 200
    - 20% of the frames in each video are corrupted

Block Dropout

Tear

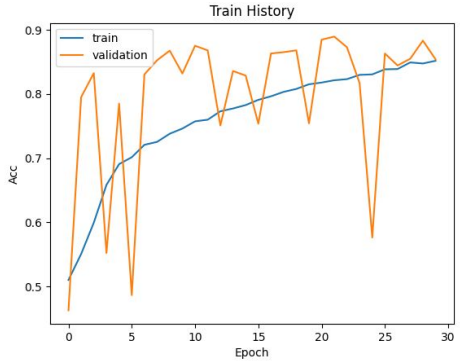
Mosaic Distortion

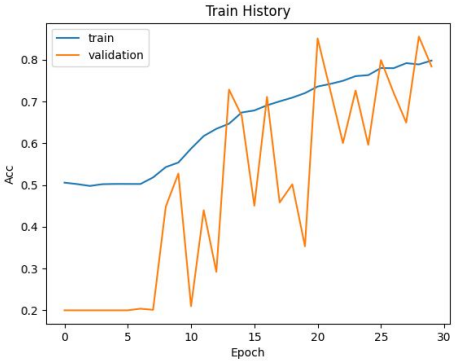


Ghosting

Before Final Results…

We have tried several model structure, different loss function design, or data augmentation to do the object detection task, all of them with backbone of ResNet18 or ResNet50 and simple head with fully connected layer or a few convolution layer; however, they finally get weird and bad result like…

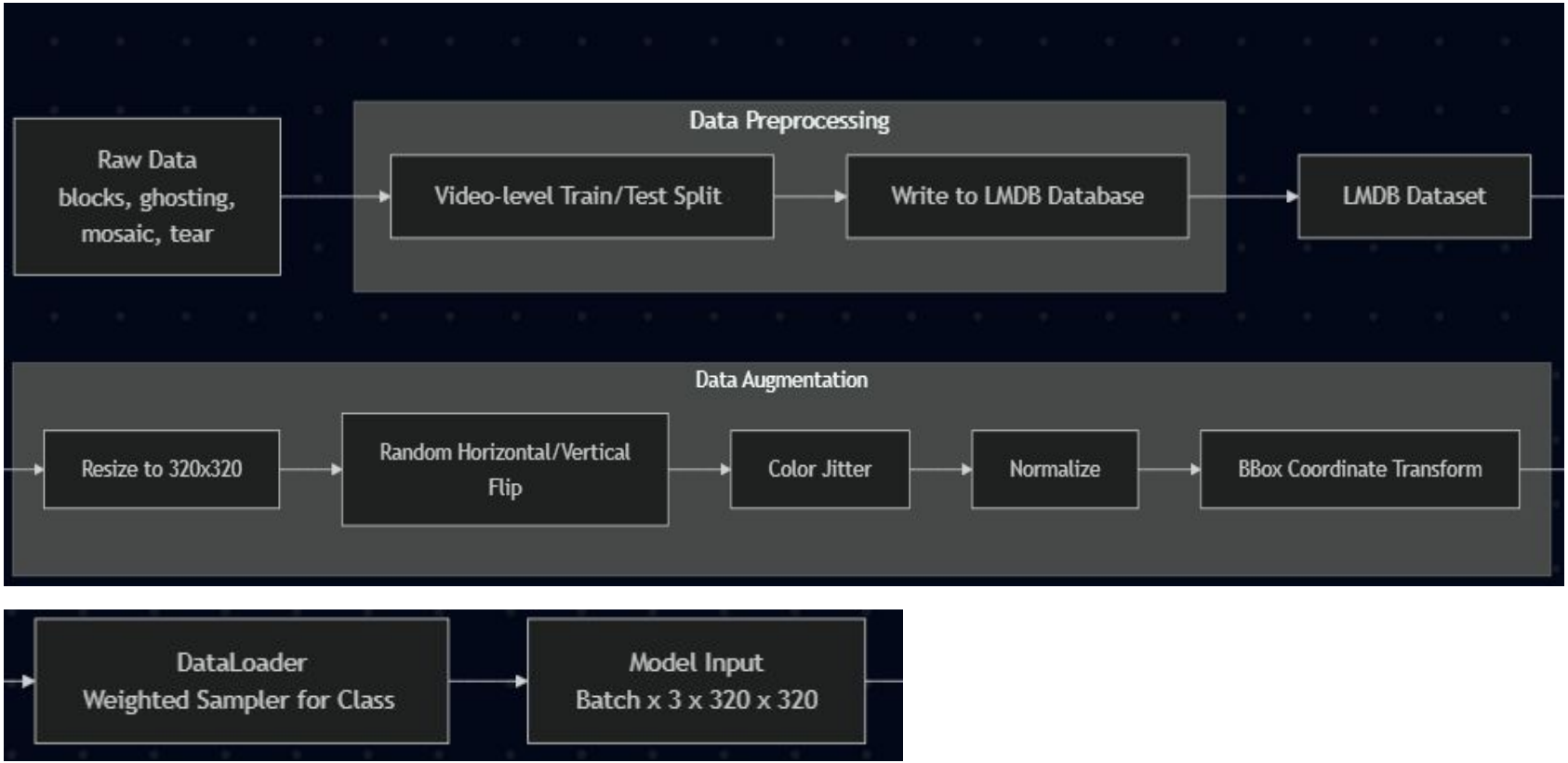




Before Final Results…

We think that the key reason of bad performance is resulted from the weak head structure. Therefore, we have proposed a model, composed of ResNet18 backbone with FPN structure (combine strong space information from shallow layer and strong semantic meanings from deep layer) and FCOS head

Proposed Framework



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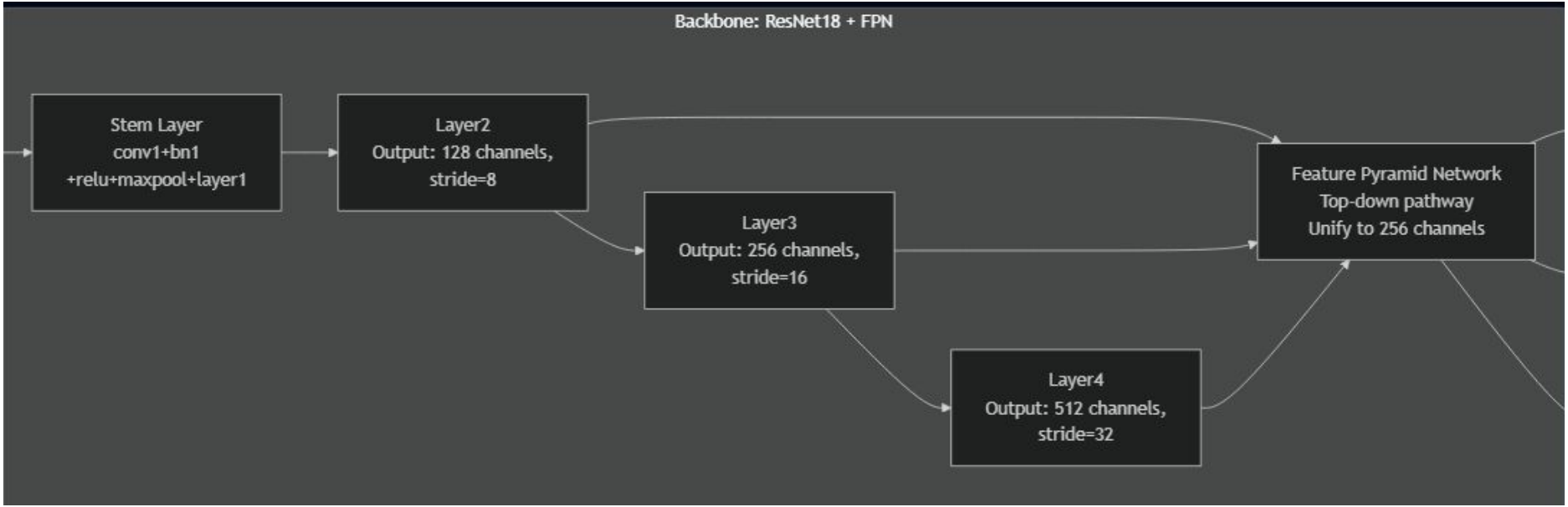

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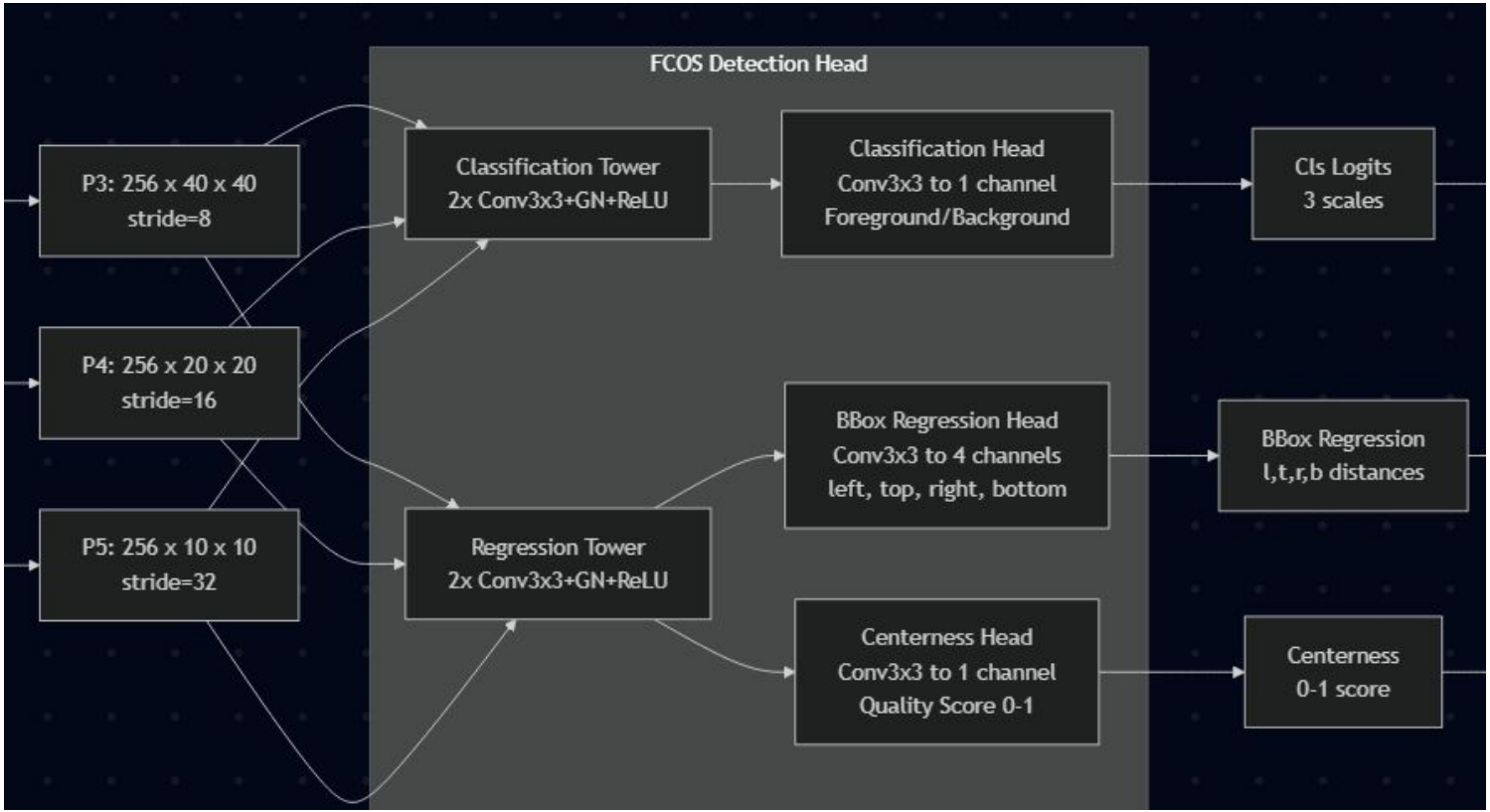

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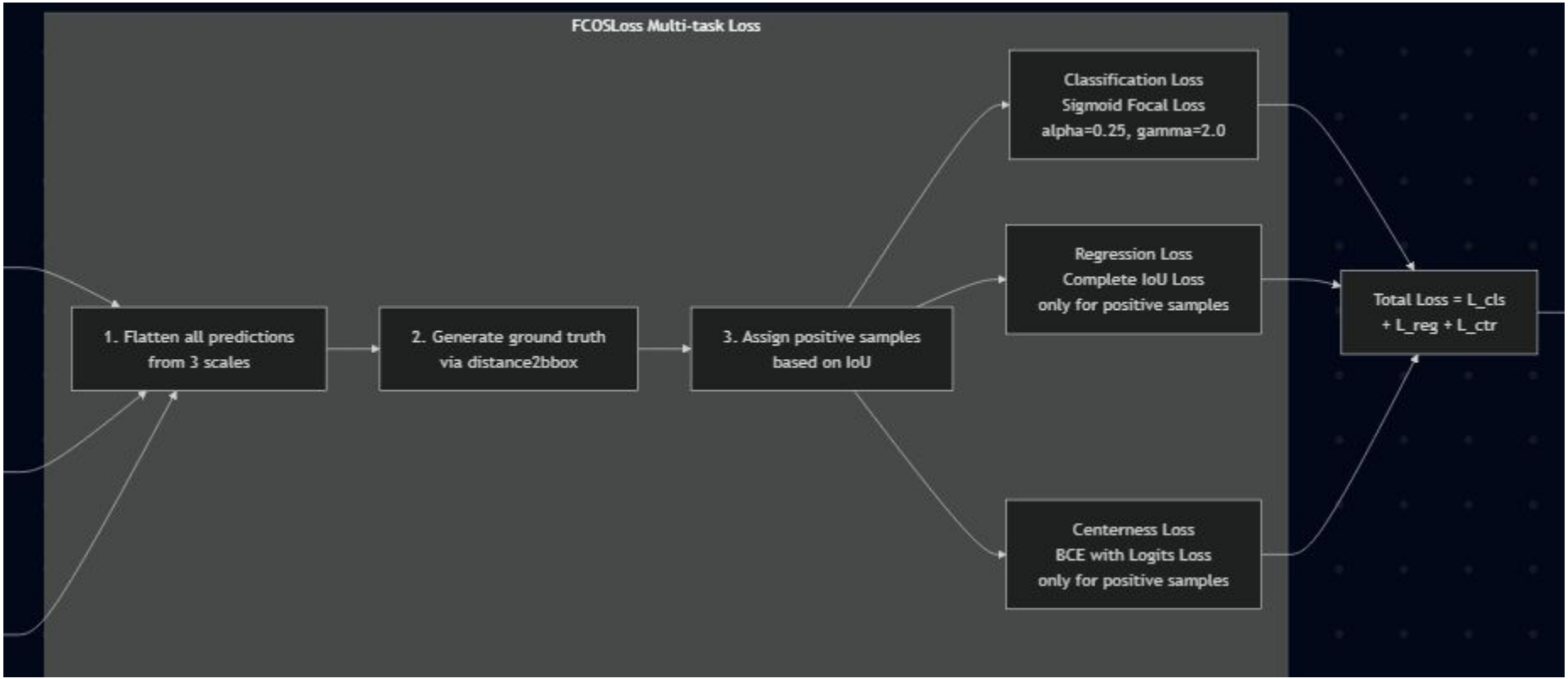
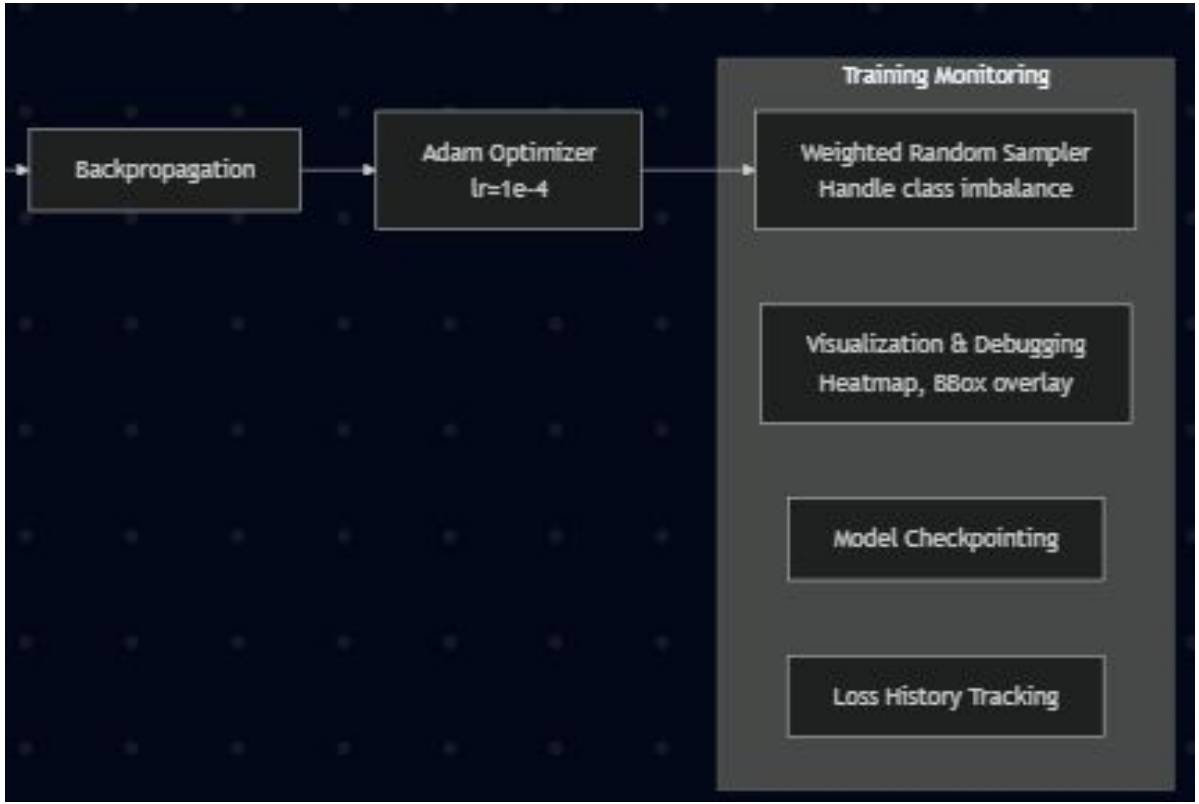


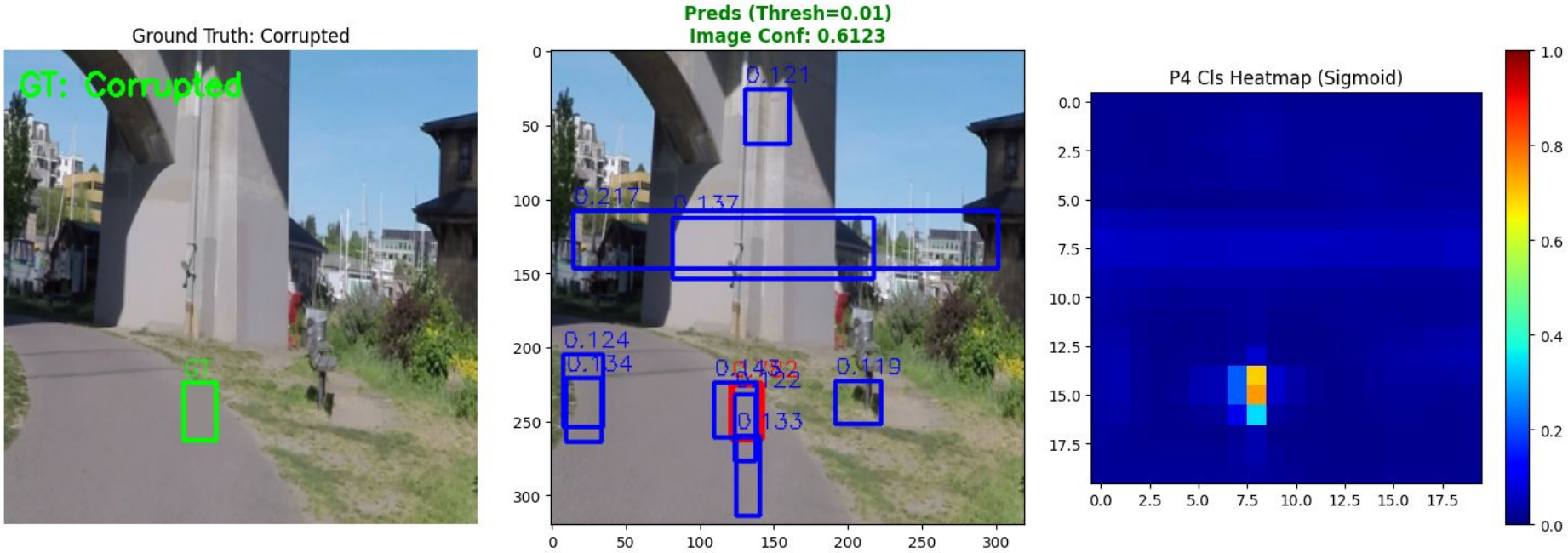
Proposed Framework



Experiments

1. data preperation
    a. build lmdb Dataset from original 4Corruption dataset
    b. split into 8:2 (train:validation)
2. configuration
    a. batch size: 64
    b. epochs: 20
    c. AMP training
3. training loop
    a. while training and validating, record accuracy, loss, IoU, mAP to observe performance
    b. write information to record.txt
    c. save best mAP and best mAP50 model

Experiments



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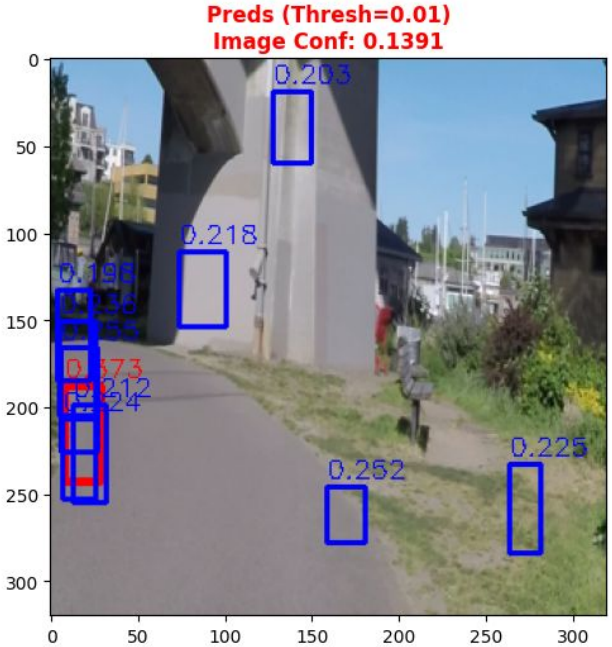

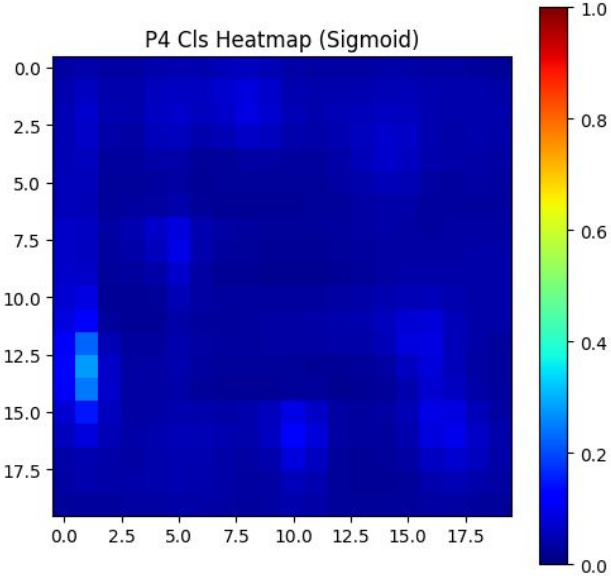

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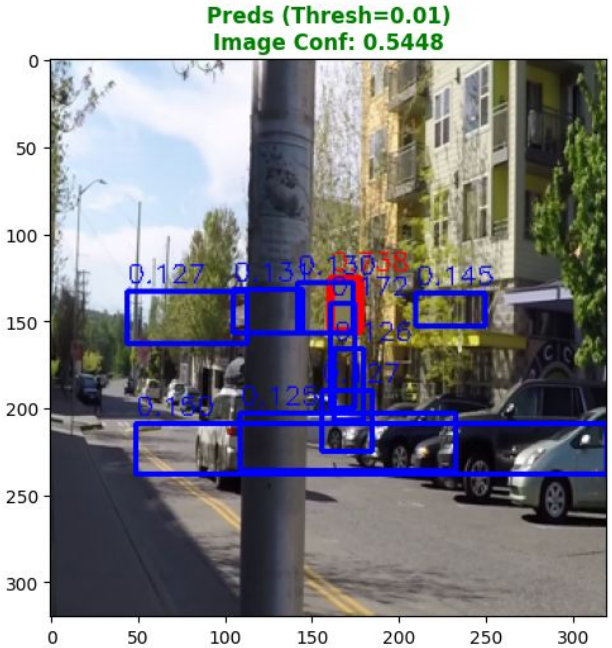

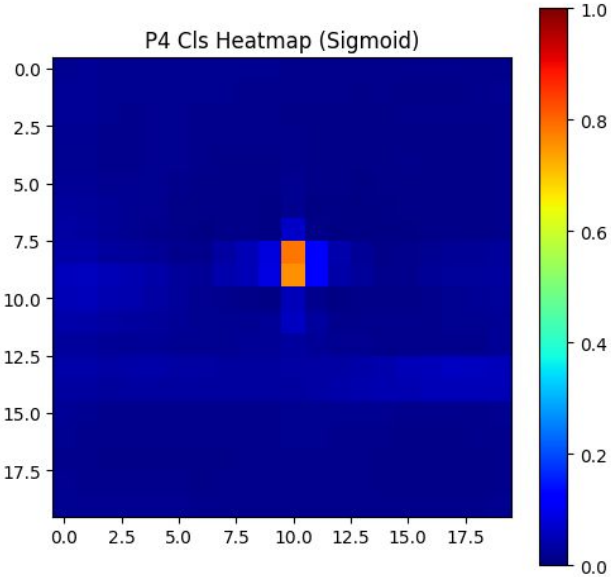

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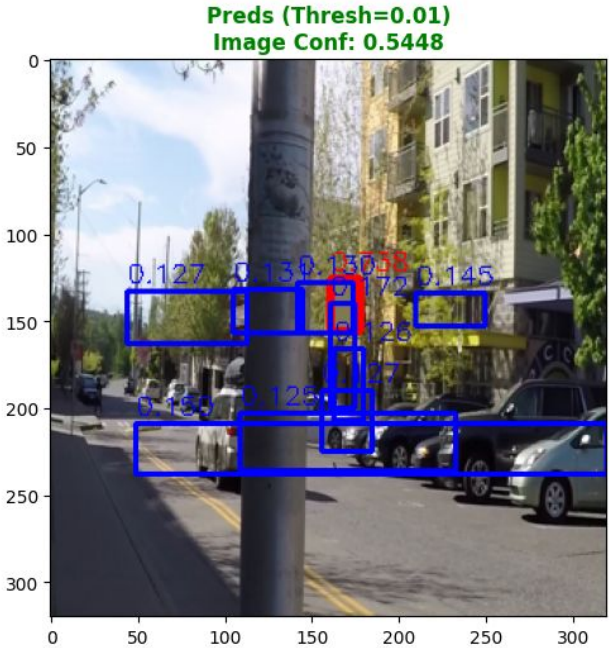

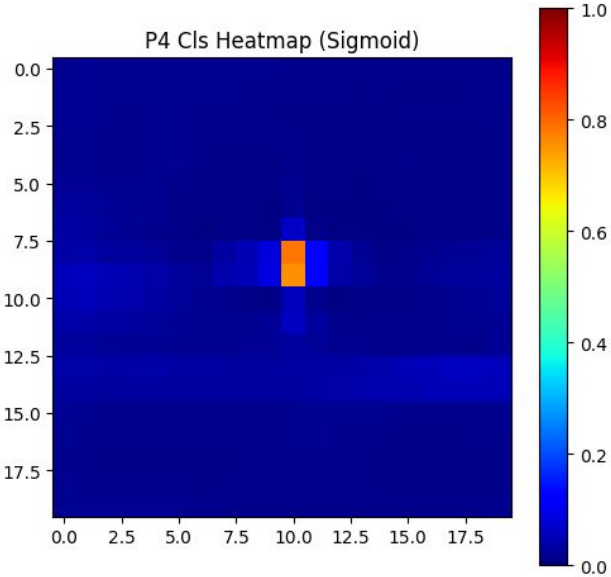

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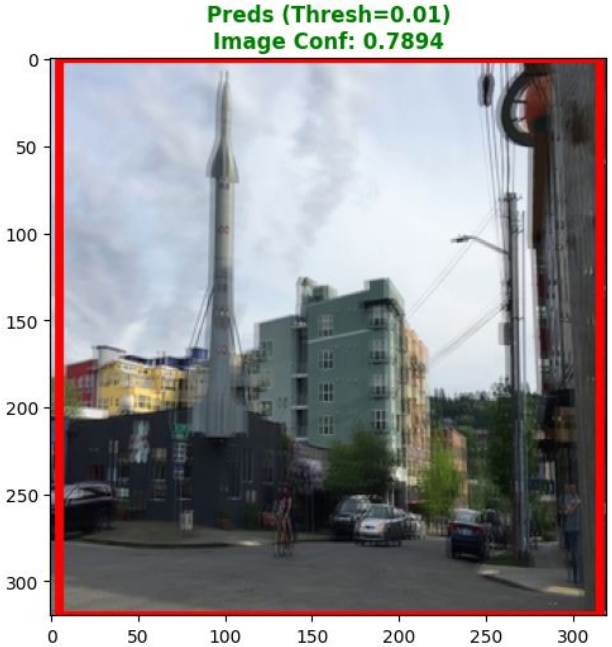

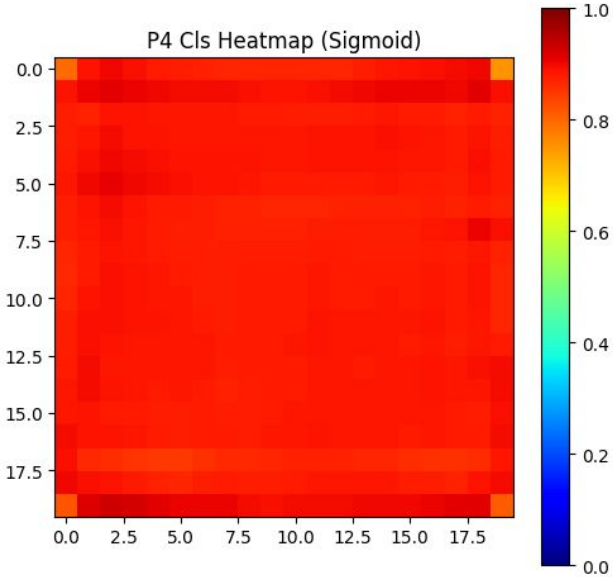

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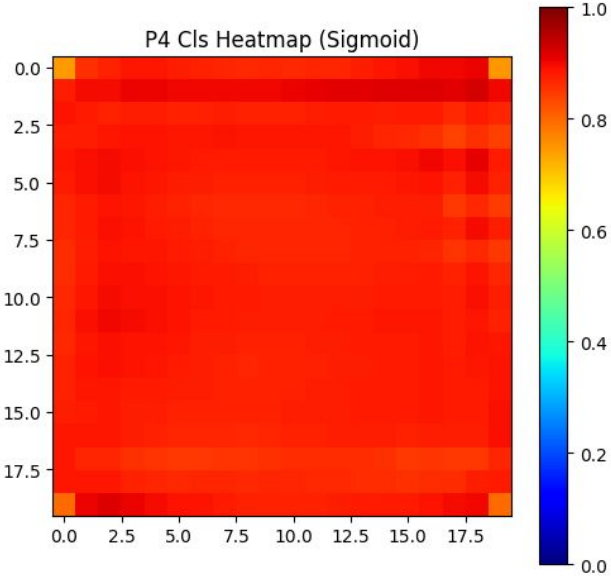

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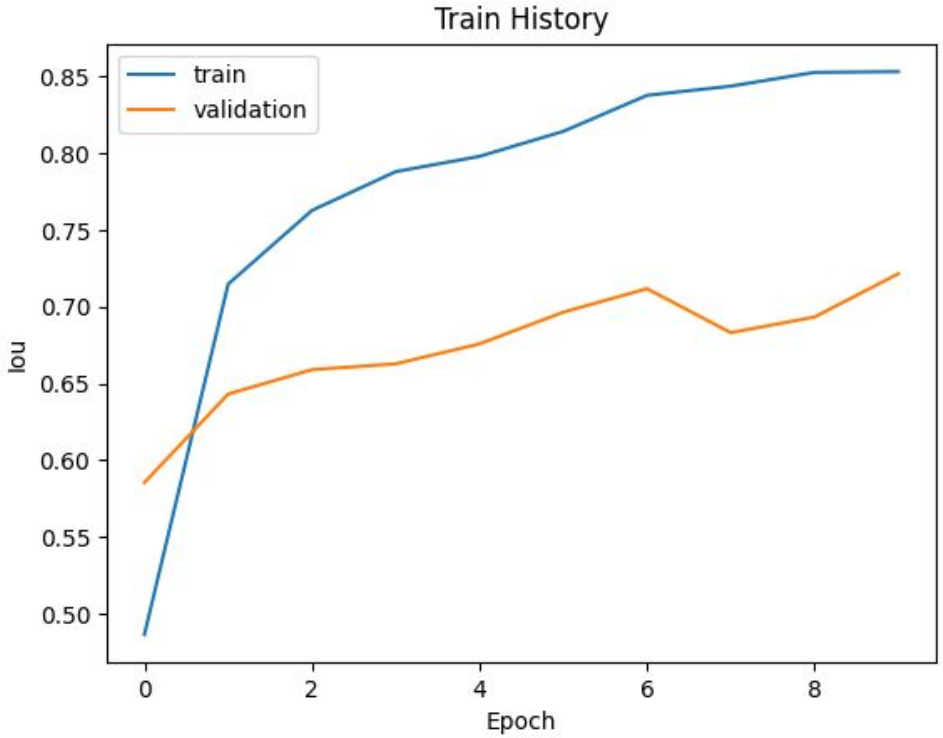
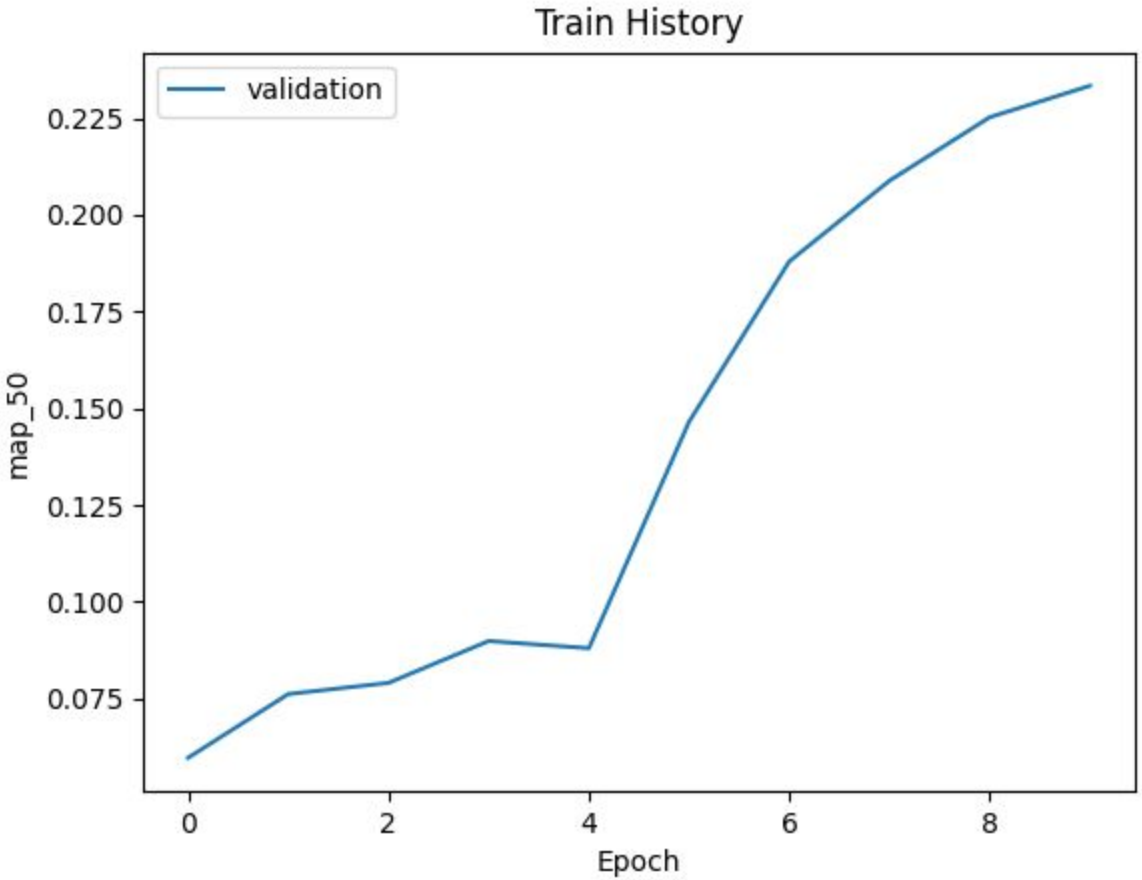
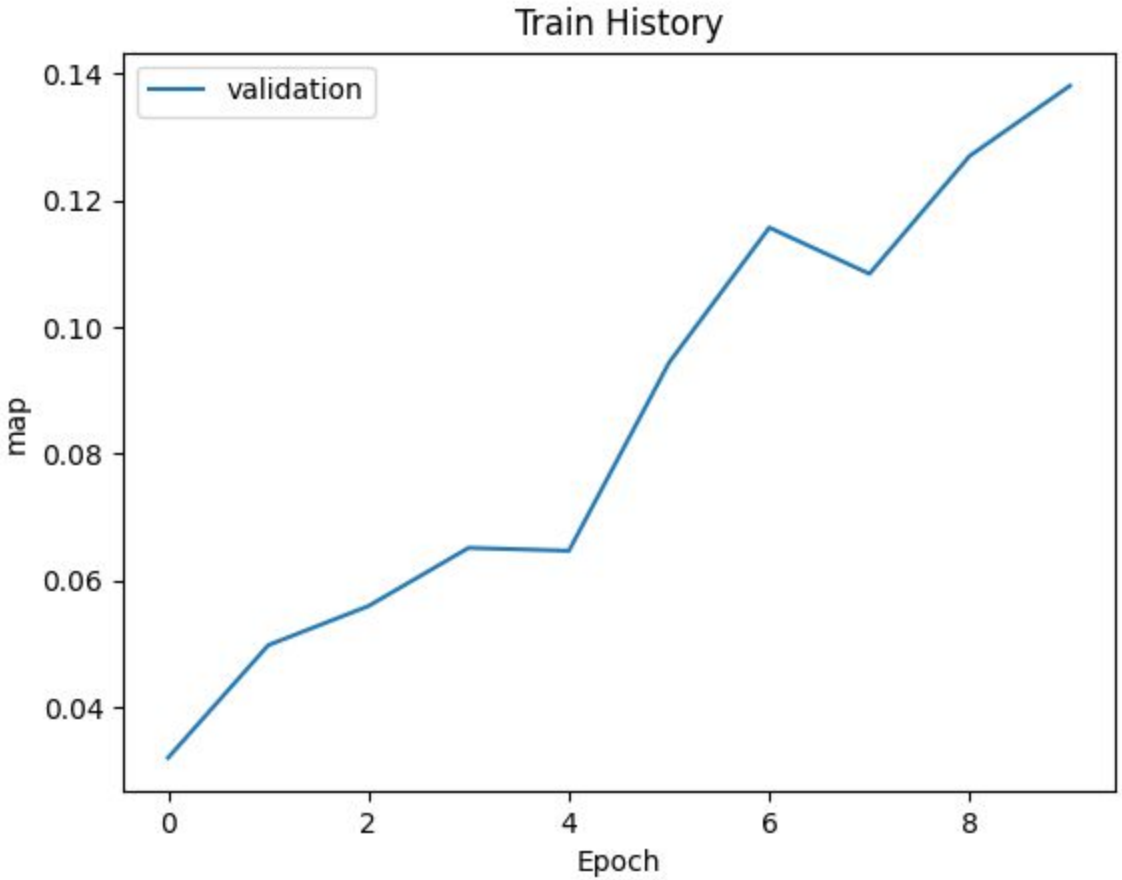
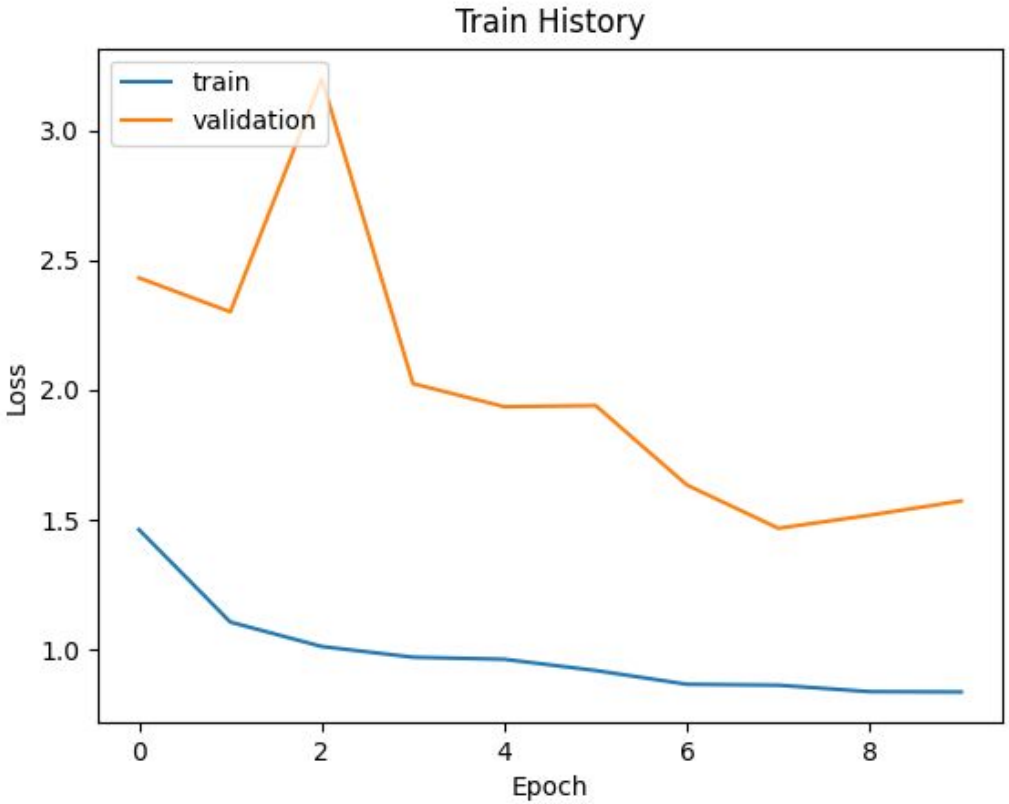
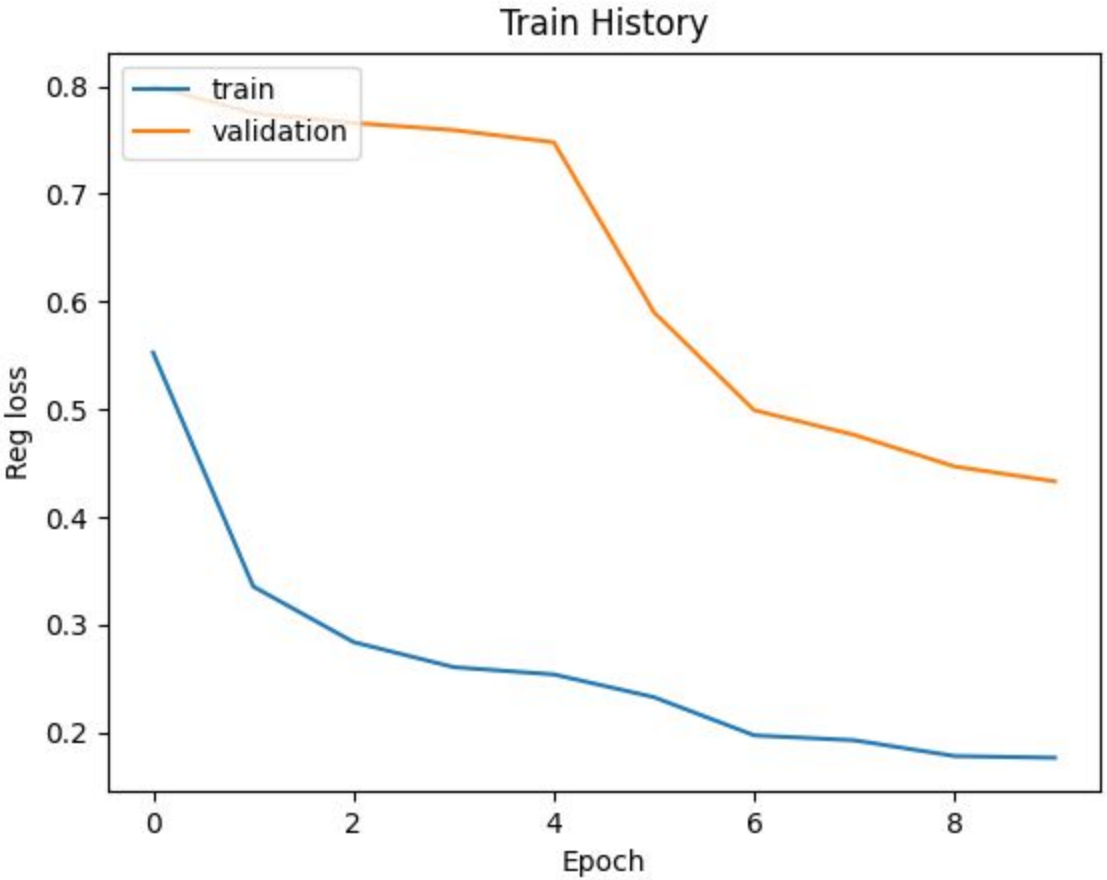
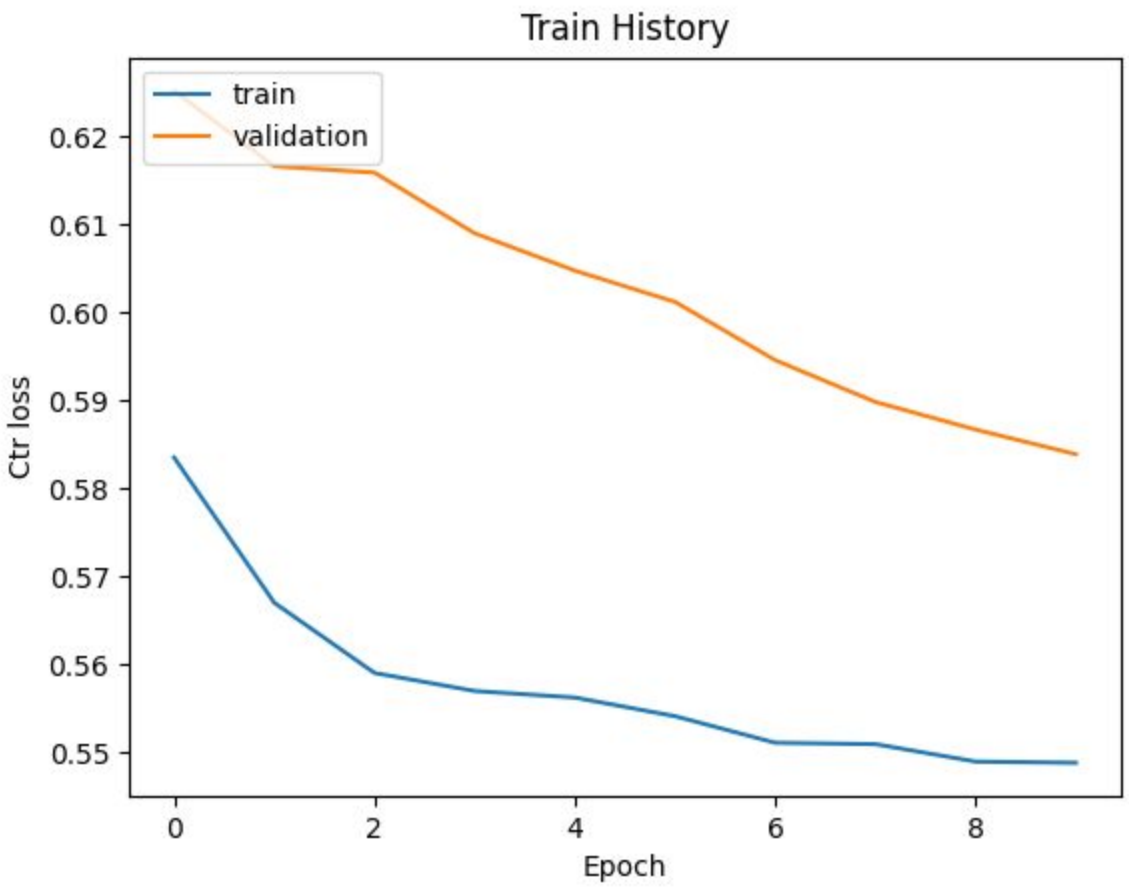


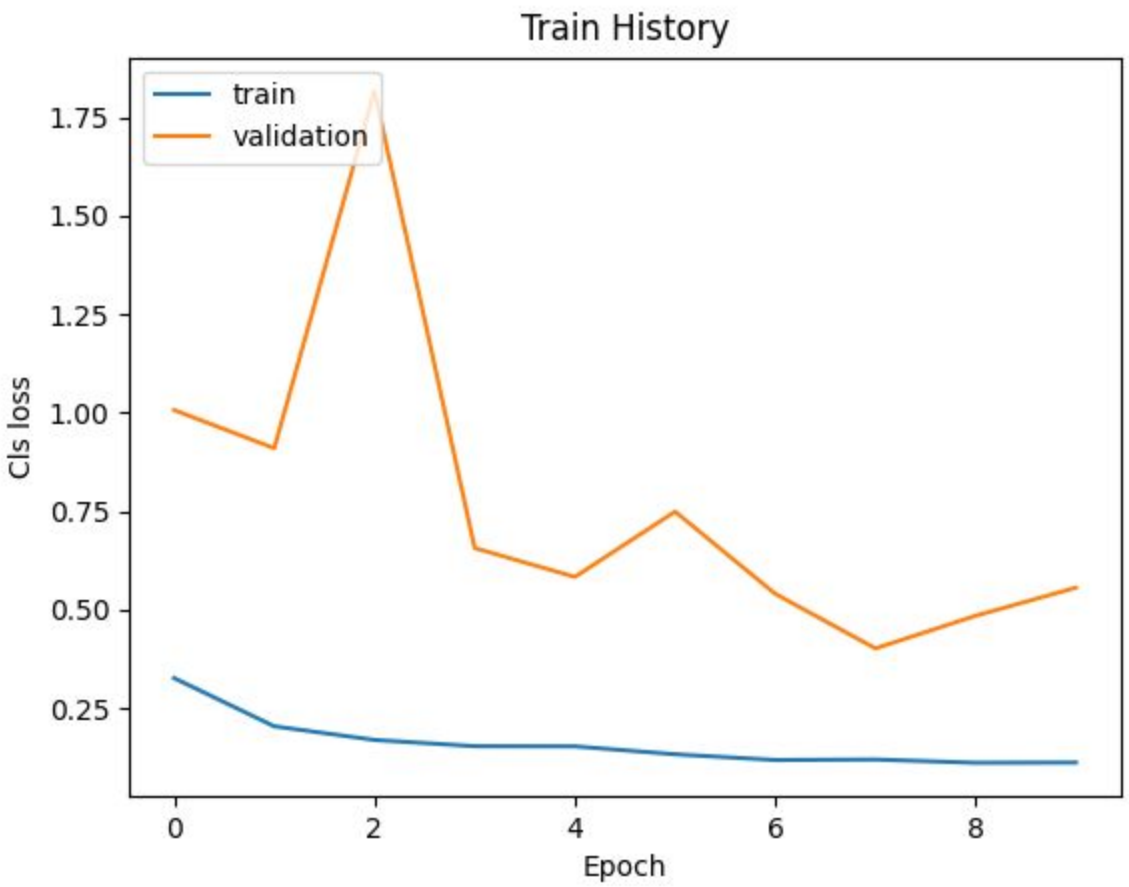

Train History
validation
0.225
0.200
0.175
0.150
0.125
0.100
0.075
map_50
0
2
4
6
8
Epoch


Train History
validation
0.14
0.12
0.10
0.08
0.06
0.04
map
0
2
4
6
8
Epoch


Train History
train
validation
3.0
2.5
2.0
1.5
1.0
Loss
0
2
4
6
8
Epoch


Train History
train
validation
Reg loss
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0
2
4
6
8
Epoch


Train History
train
validation
0.62
0.61
0.60
0.59
0.58
0.57
0.56
0.55
Ctr loss
0
2
4
6
8
Epoch


Train History
train
validation
1.75
1.50
1.25
1.00
0.75
0.50
0.25
Cls loss
0
2
4
6
8
Epoch

Conclusion

1. Best Performace under 20 epochs
   Accuray: 0.8730

   IoU: 0.7177

2. Best Performance under 50 epochs with early stopping
   Accuracy: 0.9120
   Val IoU: 0.7178